

Dynamic Edge Association and Resource Allocation in Self-Organizing Hierarchical Federated Learning Networks

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Abstract—Federated Learning (FL) is a promising privacy-preserving distributed machine learning paradigm. However, communication inefficiency remains the key bottleneck that impedes its large-scale implementation. Recently, hierarchical FL (HFL) has been proposed in which data owners, i.e., workers, can first transmit their updated model parameters to edge servers for intermediate aggregation. This reduces the instances of global communication and straggling workers. To enable efficient HFL, it is important to address the issues of edge association and resource allocation in the context of non-cooperative players, i.e., workers, edge servers, and model owner. However, the existing studies merely focus on static approaches and do not consider the dynamic interactions and bounded rationalities of the players. In this paper, we propose a hierarchical game framework to study the dynamics of edge association and resource allocation in self-organizing HFL networks. In the lower-level game, the edge association strategies of the workers are modelled using an evolutionary game. In the upper-level game, a Stackelberg differential game is adopted in which the model owner decides an optimal reward scheme given the expected bandwidth allocation control strategy of the edge server. Finally, we provide numerical results to validate that our proposed framework captures the HFL system dynamics under varying sources of network heterogeneity.

Index Terms—Federated Learning, Edge Intelligence, Resource Allocation, Evolutionary Game, Stackelberg Differential Game

I. INTRODUCTION

Recently, in view of growing privacy concerns, stringent privacy protection laws such as the General Data Protection Regulation (GDPR) have been introduced. This potentially impedes the development of Artificial Intelligence (AI) based solutions which are predominantly cloud-centric, i.e., where data is transmitted to a third-party cloud server for processing.

In response, Federated Learning (FL) [1] has emerged as a promising distributed machine learning paradigm to break down the information silos while enabling privacy preservation towards the development of AI empowered applications. In FL, model training takes place locally on each data owner's (hereinafter *worker*) device, e.g., smartphones [2] and Unmanned Aerial Vehicles [3], [4]. Then, only the updated model

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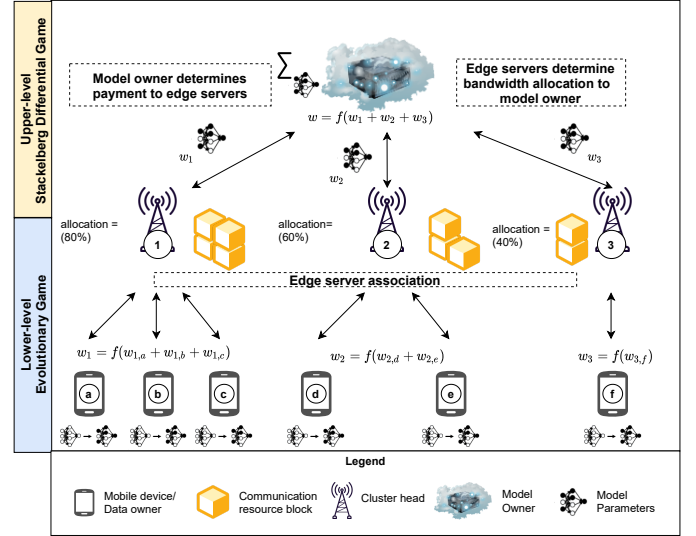


Fig. 1: The system model.

parameters, rather than the raw data, are transmitted to a model owner for weighted aggregation. During the collaborative training process, a global model is derived without the explicit exposure of a worker's raw data to third-parties.

FL networks are envisioned to comprise thousands of heterogeneous IoT and mobile devices [5]. As such, communication inefficiency remains a key bottleneck [6]. To reduce instances of node failures, solutions from cooperative communications [7] and edge computing [8] have recently been proposed to jointly address the learning, communication, and computation resource management challenges towards realizing an adaptive, scalable, and sustainable implementation of FL. In [9]–[13], the hierarchical FL (HFL) framework¹ is proposed in which workers first upload their local parameters to edge servers, e.g., small cell base stations (SBSs) in cellular networks or Roadside Units (RSU) in vehicular networks, for intermediate aggregation. The intermediate parameters are then transmitted by each of the edge servers during specified time intervals to the model owner, e.g., macro-cell base station.

There are key advantages in the HFL framework. Firstly, the number of costly global communication rounds with the remote model owner is reduced [9]. Secondly, the relay

¹Note that [11] and [14] propose the similar frameworks of self-organizing FL and collaborative FL respectively.

approach reduces the dropout rate of workers that may lack communication resources [15], [16]. Given the synchronous nature of global aggregation, the training process is inevitably delayed by straggling workers. In response, the edge server can provide communication resources, e.g., bandwidth, to facilitate efficient uplink [11]. Thirdly, the intermediate aggregation phase in the HFL framework can also be utilized to improve the personalization learning performance of FL [17]. Finally, the convergence of HFL has been proven. In addition, empirical studies do not establish a significant reduction in model accuracy, relative to the conventional FL implementation in which direct communication with the model owner is implemented [9].

In practical implementation of HFL, there exists non-cooperative players, i.e., the model owner, edge servers, and workers, all of which desire to maximize their utilities or profits while being subject to heterogeneous resource constraints and preferences. Specifically, the edge servers and model owner may be different entities and therefore, there is no incentive for the edge servers to share its resources with the model owner. To ensure scalable and efficient HFL implementation, it is therefore essential to address the open issues of edge association and resource allocation while taking into account the key unique aspects of HFL:

- *Self-organizing*: In an HFL network, the workers can autonomously decide on the edge server to associate with to improve their utilities while participating in the collaborative model training task. In addition, the edge servers can decide on the proportion of resources, i.e., bandwidth, to allocate to the model owner in support of the HFL implementation so as to maximize their respective profits. Finally, based on individual preferences, the model owner also independently decides on the payment that it accords to the edge servers for their services. In other words, the decisions of the players in the HFL network are made autonomously without the coordination of a centralized entity.
- *Bounded rationality*: The workers gradually adapt their edge association strategies to improve their utilities, whereas the edge servers and model owner gradually adjust their resource and payment allocation strategies to improve their profits respectively. In contrast to conventional optimization or game theoretic problem formulations, the strategies of the players are not immediately derived [18].
- *Dynamic*: The non-cooperative players have the objectives of utility and profit maximization in mind. To achieve their objectives, they observe each others' strategies, and formulate their decisions accordingly.

While the self-organizing aspect of HFL is considered in existing studies [11], the interactions between the workers and edge servers are often modeled to be i) static (i.e., one-shot and non-repeating) and ii) fully rational, i.e., a steady state of the network formation and resource allocation strategy is reached immediately, rather than derived dynamically. However, the above assumptions do not hold in reality. As a result, inefficient resource allocation occurs if the strategies of the

players are devised with the assumption that the system state is already at an equilibrium after one-shot interactions.

In this paper, we consider the HFL system model with non-cooperative players. To facilitate efficient uplink of model updates, the model owner employs the services of edge servers to provide communication resources and relay support for the workers. Each worker in turn chooses an edge server to associate with. In consideration of the dynamics of edge association and resource allocation, we consider a hierarchical game framework (Fig. 1) as follows:

- *Lower-level evolutionary game*: The workers select the edge server to associate with through an evolutionary game approach. We capture the dynamics of edge server association with the replicator dynamics.
- *Upper-level stackelberg differential game*: We address two problems in the upper-level Stackelberg differential game. Firstly, as the leader, the model owner has to devise the optimal amount of payment to accord to the edge servers for their resources. For example, if a model owner requires the task to be completed more quickly, or if more workers are required, the model owner is willing to pay more. Secondly, as the followers, the edge servers select the optimal proportion of bandwidth to allocate to the model owner as their best responses to the model owner's payment strategy.

The main contributions of this paper are as follows:

- 1) We propose a hierarchical game framework to model the self-organizing, bounded rationality, and dynamic aspects of edge association and resource allocation in HFL. These aspects capture the interdependence of interactions among the model owner, edge servers, and workers in a self-organizing HFL network such that efficient resource allocation is achieved.
- 2) We model the edge association decisions of the workers as an evolutionary game. Contrary to existing studies, our paper addresses the dynamic and bounded rationality aspects in network formation to capture holistically the interactions among workers during the edge association process.
- 3) Using the Stackelberg differential game formulation, we derive the open-loop Stackelberg equilibrium as a solution to capture the dynamics of non-cooperative model owner's and edge servers' interactions. The nature of the open-loop equilibrium is such that it may be derived even without the intervention of a central controller, thus reducing communication bottlenecks and enhancing the scalability of HFL networks. In addition, the dynamic game formulation maximizes the accumulated profits of the players in the game over static formulations as a result of efficient resource allocation.
- 4) For performance evaluation, we validate the robustness of our proposed framework to varying levels of resource heterogeneity that reflects the practical considerations towards the actual implementation of HFL.

The rest of this paper is organized as follows. In Section II, we present a review of the related works. In Section III, we present the system model. In Section IV, we discuss the

evolutionary game based edge association and network formation. In Section V, we propose the Stackelberg differential game framework for dynamic resource allocation. Finally, we conduct the performance evaluation in Section VI and conclude in Section VII.

II. RELATED WORKS

FL is a privacy-preserving machine learning paradigm first proposed in [1]. In distributed learning schemes such as FL, the communication cost often dominates the computation cost. In particular, the uplink transmission rate of workers is a major bottleneck in the training process that can lead to the straggler's effect [2]. Several works have proposed a variety of solutions, e.g., model compression techniques such as quantization and subsampling [19], client selection protocols to reduce the occurrences of stragglers [20], robust design to reduce the effect of noise for FL over noisy channels [21], [22] and to improve energy efficiency [23], as well as Broadband Analog Aggregation (BAA) with over-the-air computation to enable the reuse of the whole bandwidth for scalable FL [24].

However, despite the above measures, the FL process is still susceptible to device dropouts and stragglers due to communication inefficiencies [25]. In addition, workers that are located at geographically distant locations are unable to participate in the FL model training process [26], [27]. This affects the model's ability to generalize well [6], [17]. Recently, edge computing-inspired solutions [28], [29] have been proposed to bring intelligence to the wireless edge [30], [31]. The work in [32] is the first to propose a heterogeneous MEC framework that achieves the global-optimal latency performance. It also complements the limitations of existing MEC works by considering the recovery time minimization in the network blocking state, particularly to guide latency-sensitive mobile communications and computing [33]. To further enhance the communication efficiency of FL, edge-computing inspired solutions have similarly been proposed. These methods generally attempt to reduce the reliance of the FL training process on a central controller, i.e., the distant cloud server of the model owner. In [9], the HFL framework is proposed in which the workers do not communicate directly with a central controller, i.e., the model owner. Instead, the intermediate aggregation of parameters is first conducted at the edge, e.g., with the aid of edge servers, such as, base stations or RSUs. Then, communication with the central controller is established only when there is a need for global aggregation.

Similarly, [11] proposes that mobile devices can form self-organized clusters to facilitate the cooperative communications and transmission for efficient FL. Besides improving communication efficiency, it reduces the likelihood that the training fails due to the unexpected malfunctioning of the central controller. In [17], it is proposed that the intermediate aggregation phase in the HFL framework can be utilized to improve the personalization performance of FL. Specifically, end users that share more common characteristics, e.g., in terms of closeness of preferences in building a recommender system, may benefit more if they are grouped together. The base layers of the neural network (for generalization) can be

trained using conventional federated averaging at the servers of the model owner, whereas the deeper layers of the neural network (for personalization) can be trained at the edge.

In this regard, HFL has emerged as a promising solution that enables the adaptive, scalable, and sustainable implementation of FL by leveraging on the resources at the network edge. However, to efficiently conduct HFL, it is of paramount importance to consider resource allocation schemes that account for the network heterogeneity. In [11], the edge server selection algorithm is proposed, and the edge server is chosen based on its social relationship with the other devices. In [10], [14], the collaborative FL framework is proposed in which the device-to-device (D2D) and device-to-edge (D2E) communication are leveraged to ensure the efficient transmission of model parameters to the model owner. In addition, various networking topologies are proposed to fulfill different FL training tasks [14]. However, while the self-organizing aspect of HFL is considered in the above studies, the interactions among the workers and edge servers are often modelled to be static (i.e., one shot and non-repeating) and fully rational, i.e., a steady state of the network is reached immediately, rather than derived dynamically, i.e., the steady state is derived after some iterations of learning (based on the learning rate), which deviates from reality. In practical implementation of HFL, it is to be expected that the workers and edge servers both have time-varying decisions. As such, static models are unable to accurately capture the resulting system dynamics.

In contrast, bounded-rationality driven network formation in self-organizing heterogeneous wireless networks have been actively studied in [18], [34], whereas dynamic resource allocation has been studied in [35], [36] to better capture the dynamics of player interactions towards deriving the equilibrium in practical networks. Inspired by these studies, we propose a hierarchical game framework that captures the self-organizing, bounded rationality, and dynamic aspects of player interactions in the HFL networks.

Furthermore, most of the existing studies on resource allocation in HFL consider only the model owner-edge server interaction, or the edge server-worker interaction [11]–[13]. In practice, the decisions of all the aforementioned players are coupled and interdependent. In this regard, our hierarchical game framework addresses the interdependence of the non-cooperating players.

III. SYSTEM MODEL

We consider a network that consists i) a model owner, ii) N workers represented by the set $\mathcal{N} = \{1, \dots, n, \dots, N\}$, and iii) J edge servers represented by the set $\mathcal{J} = \{1, \dots, j, \dots, J\}$. The model owner aims to train a model, with parameters denoted by a vector $\mathbf{w}$, over K iterations to minimize the global loss $L^K(\mathbf{w})$. The FL training for the model owner is conducted collaboratively with the contribution of trained parameters from the N workers. Note that the loss function can in turn be calibrated to accomplish the different learning tasks required by different model owners.

Given the communication constraints of individual workers, the cloud-centric conventional FL architecture is prone to

high device dropout rates [14]. Therefore, the HFL system architecture (Fig. 1) is considered in which edge servers, e.g., base stations in cellular networks or RSUs in vehicular networks, are employed across the network to facilitate the uplink transmission of parameter updates through distributing orthogonal resource blocks to its workers. Then, each worker chooses an edge server to associate. The HFL iteration k consists of three main steps as follows [19].

- 1) *Local Computation*: Each worker² n downloads the intermediate model from the edge server j denoted by $w_j^{(k)}$ and trains the model locally. For clarity, we refer to the model aggregated across all edge servers as the *global* model, the model aggregated across all workers belonging to a edge server as the *intermediate* model, and the model derived after local training as the *local* model.
- 2) *Wireless Transmission*: Each worker n transmits the updated local model $w_{n,j}^{(k)}$ to its edge server j .
- 3) *Model Parameter Aggregation*: The edge server j aggregates all local parameters from its workers, e.g., through weighted averaging in *FedAvg* [1], to derive an updated intermediate model $w_j^{(k+1)}$. At predefined intervals $\tilde{k}$, all edge servers upload $w_j^{(k+\tilde{k})}, \forall j \in \mathcal{J}$, to the model owner for aggregation to derive the updated global model $w^{(k+\tilde{k})}$. Otherwise, the intermediate model $w_j^{(k+1)}$ is transmitted back to the workers for the $(k+1)^{th}$ training iteration.

In this paper, we assume that the J edge servers are pre-determined, e.g., through selection algorithms that are based on energy efficiency [37], [38], trust [39], or social effects [40]. We focus our study on a hierarchical game formulation as follows:

- 1) *Lower-level Evolutionary Game*: In the lower-level, we study the edge association strategies of the workers. The HFL network is self-organizing, i.e., the workers autonomously decide which edge server to join. Their decisions are in turn characterized by *bounded rationality* and *dynamic interactions*. The former implies that an equilibrium solution is not attained immediately, i.e., the workers may switch from one strategy to another to seek for higher utilities before an equilibrium solution is achieved. For the latter, the decisions of workers are highly interdependent upon one another, given that the workers have to share the limited resources that an edge server can provide. Therefore, a worker has to observe the actions and payoffs of other workers before making a decision [18]. In this regard, we adopt and formulate the evolutionary game model in Section IV to model the evolution of the workers' edge server association strategies.
- 2) *Upper-level Differential Game*: In the upper-level, we study the resource allocation strategies of the edge servers, and the reward allocation strategy of the model

owner. Given that the model owner has different preferences, i.e., in terms of its valuation for bandwidth contributed by the edge server, the model owner will devise a suitable reward allocation strategy to maximize its present value profits. In response, each edge server will choose to allocate varying amount of its bandwidth to the model owner, in its objective to maximize its present value profits. To model the dynamic interactions between the model owner and edge servers, we adopt the Stackelberg differential game approach in Section V.

IV. LOWER-LEVEL EVOLUTIONARY GAME

In this section, we adopt the evolutionary game to capture the bounded rationality and dynamic aspects of the workers' edge server association strategies in a self-organizing HFL network. In Section IV-A, we introduce the worker utilities and replicator dynamics that characterize the evolution in edge association strategies. We prove the existence, uniqueness, and stability of the replicator dynamics in Section IV-B.

A. Problem Formulation

We list the definitions of various aspects of the evolutionary game based edge association as follows:

- *Players*: Each $n \in \mathcal{N}$ worker is the player of the evolutionary game.
- *Population*: We have the set $\mathcal{N}$ of N workers to be defined as one population.
- *Strategy*: Each worker devises an edge association strategy, i.e., by choosing an edge server $j \in \mathcal{J}$ to associate with, so as to achieve utility maximization. We further define the utility of each worker in (3).
- *Population share and population state*: We denote $x_j(t)$ as the population share, i.e., proportion of workers that associates with edge server j at time t , where $\sum_{j=1}^J x_j(t) = 1$. The population state, i.e., proportion of the population that has joined edge server $j \in \mathcal{J}$, is denoted by $\mathbf{x} = [x_1, \dots, x_j, \dots, x_J]^T \in \mathbb{X}$. Note that $\mathbb{X}$ refers to the state space that consists of all feasible population distributions. Note that t may refer to each training iteration or instance of FL model training, i.e., after the global model is aggregated, the strategy of the worker may be adjusted.

Each edge server j has R_j available radio resource blocks. The edge server chooses to allocate $\zeta_j(t)R_j$ resource blocks to support the FL training for the workers, where $\zeta_j(t) = [0, 1]$. Note that the edge servers' preferences of $\zeta_j(t), \forall j \in \mathcal{J}$, is further explored as a dynamic resource allocation problem to maximize the edge server's present value profits in response to the expected reward allocation strategy of the model owner in Section V. The edge servers' resource allocation strategies are denoted by the vector $\boldsymbol{\zeta} = [\zeta_1, \dots, \zeta_j, \dots, \zeta_J]^T$. Note that we use $\zeta_j(t)$ and ζ_j , as well as $x_j(t)$ and x_j interchangeably where dropping the time notations does not lead to confusion.

²In the case where not all workers participate in training, worker selection schemes [20] can be considered to choose just a subset of suitable workers, e.g., based on their probabilities of task completion.

Following [11], [36], each worker n under edge server j is equally allocated $r_j(\mathbf{x}(t), \zeta_j(t))$ resource blocks following the unweighted round-robin scheme [41], [42] where:

$$r_j(\mathbf{x}(t), \zeta_j(t)) = \frac{\zeta_j(t) R_j}{x_j(t) N}. \quad (1)$$

This assumption is appropriate since all workers $n \in \mathcal{N}$ are participating towards training the same global model, i.e., they have identical delay-tolerance [43]. Accordingly, the uplink rate [44] for the worker under edge server³ j is as follows:

$$B_j(\mathbf{x}(t), \zeta_j(t)) = r_j(\mathbf{x}(t), \zeta_j(t)) W \mathbb{E}_{h_j} \left(\log_2 \left(1 + \frac{P h_j}{W N_0} \right) \right), \quad (2)$$

where W refers to the bandwidth of each resource block, P refers to the worker transmit power, N_0 refers to the noise power spectral density, and $\mathbb{E}_{h_j}(\cdot)$ refers to expectation with respect to the channel gain h_j between the worker and edge server j . Our model can easily be extended to consider the case of multiple worker populations, e.g., to account for χ categories of worker heterogeneities in terms of transmission parameters, we can simply consider a total of χ distinct worker populations. We present this extension in Section VI.

For ease of presentation, we represent the terms $B_j(\mathbf{x}(t), \zeta_j(t))$ and $r_j(\mathbf{x}(t), \zeta_j(t))$ with $B_j(t)$ and $r_j(t)$ interchangeably. At time t , the utility derived by a worker for associating with edge server j is:

$$u_j(B_j(t)) = \alpha \mathcal{U} \left(\frac{S(\mathbf{w}_{n,j}^{(k)})}{B_j(t)} \right)^{-1} = \alpha \mathcal{U} \left(\frac{B_j(t)}{S(\mathbf{w}_{n,j}^{(k)})} \right), \quad (3)$$

where α is a preference parameter, e.g., the parameter dictates the disutility gain from each unit of time expended by the worker on the uplink transmission. This may be derived historically, e.g., from user studies [45]. $\mathcal{U}(\cdot)$ is the utility function indicating the risk attitude of the worker, and $S(\mathbf{w}_{n,j}^{(k)})$ refers to the size of model parameters to be uploaded. Specifically, we assume risk neutrality of the workers without loss of generality, i.e., $\mathcal{U}(\cdot)$ is a linear utility function, following the works of [3], [45], [46]. Moreover, the model parameter size to be transmitted is constant across the same population since the global model to be trained is the same regardless of the workers or iterations, i.e., $S(\mathbf{w}_{n,j}^{(k)}) = S$, $\forall k \in \{1, \dots, K\}$ and $\forall n \in \mathcal{N}$. Intuitively, when more time is required for the uplink transmission of model parameters, the worker derives lower utility. For example, this is due to greater device energy expense incurred during the delayed uplink transmission⁴ or because the device cannot be used by the worker for leisure activities in the interim [5], [6].

³Similar to [11], each orthogonal resource block allocated for the FL training task is occupied by at most one FL worker, i.e., other FL workers or users not participating in the FL training thus do not cause interference. For tractability, we do not consider interference among the edge servers. However, the presence of interference is similar to that of heterogeneous network conditions, which is explored in Fig. 10 and 11.

⁴We consider the uplink transmission in this paper due to the communication bottleneck, which is as a result of asymmetry in data transmission rates, i.e., the downlink is typically greater than the uplink.

The information regarding the utility derived by each worker from associating with different edge servers may be communicated among workers within the HFL network. Using this information, a worker may adjust its edge server association strategies. A change in strategy influences the utilities of workers across the network and thus, the association strategies of other workers. To model the evolutionary dynamics of edge association strategies for a representative population m , we consider a series of ordinary differential equations with the initial condition $\mathbf{x}(0) \in \mathbb{X}$, otherwise known as the replicator dynamics [18]:

$$\dot{x}_j(t) = f(\mathbf{x}(t), \zeta(t)) = \delta x_j(t) (u_j(B_j(t)) - \bar{u}(B_j(t))), \quad \forall j \in \mathcal{J}, \forall t, \quad (4)$$

where δ refers to the learning rate of the worker. Note that the historical learning rate can be validated empirically. In some cases, the learning rate may be higher due to stronger social ties [47] across workers of the same population. Otherwise, the learning rate may be lower due to the difficulty of obtaining information within the network [18]. The average utility of workers across the HFL network is defined as follows:

$$\bar{u}(B_j(t)) = \sum_{j=1}^J x_j(t) u_j(B_j(t)). \quad (5)$$

Specifically, based on (4), the population state $\mathbf{x}(t)$ evolves when workers compare their utilities derived as a result of their current edge association strategies to the average utilities of workers of the same population (that may have adopted other strategies by choosing the different edge servers). If the strategy of associating with an edge server j offers higher pay-offs than that of the average utility, i.e., $u_j(B_j(t)) > \bar{u}(B_j(t))$, the population share $x_j(t)$ increases since more workers churn and select the corresponding edge server, i.e., $\dot{x}_j(t) > 0$.

The evolutionary equilibrium for the population m is reached only when $\dot{x}_j(t) = 0, \forall j \in \mathcal{J}$. At the evolutionary equilibrium, all workers within the population derive the same utility, i.e., $u_j(B_j(t)) = \bar{u}(B_j(t)), \forall j \in \mathcal{J}$. The reason is that the workers within the population are no longer incentivized to alter their edge server association strategies. However, the evolutionary dynamics are in turn influenced by the bandwidth allocation control strategies devised by the edge servers. In this regard, we prove the existence, uniqueness, and stability of the solution to (4) in the next section.

B. Existence, Uniqueness, and Stability Analyses of the Evolutionary Equilibrium

Next, we first prove the uniqueness and existence of the solution to (4), i.e., the evolutionary equilibrium.

Theorem 1. *If every element ζ_j in the vector ζ is measurable on $[0, \infty)$, there exists a unique solution $\mathbf{x}(t)$ to the replicator dynamics in (4) with the initial population state $\mathbf{x}(0) = \bar{\mathbf{x}}$.*

Proof. For fixed t , the partial derivative of $f(\mathbf{x}(t), \zeta(t))$ with respect to $x_j(t)$ is continuous. Given that $\zeta_j(t)$ is measurable on $[0, \infty)$, this implies that $f(\mathbf{x}(t), \zeta(t))$ is measurable for a fixed $x_j(t)$ on $[0, \infty)$. Furthermore, for any given closed

bounded set $\psi \in \mathbb{X}$ and the interval $[a, b] \in [0, \infty)$, there always exists a positive number v in which an integrable function can be constructed where:

$$D_j(t) = \left| \frac{\frac{\zeta_j R_j}{N} \delta \alpha W \mathbb{E}_{h_j} \left(\log_2 \left(1 + \frac{P_{h_j}}{W N_0} \right) \right)}{S} \right| + v \left| \sum_{j=1}^J \frac{\frac{\zeta_j R_j}{N} \delta \alpha W \mathbb{E}_{h_j} \left(\log_2 \left(1 + \frac{P_{h_j}}{W N_0} \right) \right)}{S} \right| \quad (6)$$

such that the inequalities $|f(\mathbf{x}(t), \zeta(t))| \leq D_j(t)$ and $\left| \frac{\partial f(\mathbf{x}(t), \zeta(t))}{\partial x_j(t)} \right| \leq D_j(t)$ holds for all $(\mathbf{x}, t) \in \psi \times [a, b]$.

Moreover, we denote $\hat{D} = \sum_{j=1}^J \frac{\frac{\zeta_j R_j}{N} \delta \alpha W \mathbb{E}_{h_j} \left(\log_2 \left(1 + \frac{P_{h_j}}{W N_0} \right) \right)}{S}$.

It follows that $|f(\mathbf{x}(t), \zeta(t)) - f(\mathbf{y}(t), \zeta(t))| = \hat{D} |\mathbf{x} - \mathbf{y}|$. Therefore, we can define a relation in which:

$$|f(\mathbf{x}(t), \zeta(t)) - f(\mathbf{y}(t), \zeta(t))| \leq L |\mathbf{x}(t) - \mathbf{y}(t)|, \quad (7)$$

where $L = \hat{D} + 1$. Note that (7) implies that the global Lipschitz condition is satisfied, and consequently, the solution to the replicator dynamics exists globally and is unique. $\square$

With Theorem 1, the existence and uniqueness of the evolutionary equilibrium is validated. Therefore, we are able to derive the evolutionary equilibrium by solving (4) when $\dot{x}_j(t) = 0, \forall j \in \mathcal{J}$. At the evolutionary equilibrium, all the workers within the population have the same utilities and thus, they do not alter their edge server association strategies.

The evolutionary stable strategy (ESS) is the steady state strategy achieved through the evolutionary process in which successful strategies spread over time. There exist two types of evolutionary equilibriums as solutions to the replicator dynamics in (4), i.e., i) the *boundary* solution, and ii) the *interior* solution. For the former, we have $x_j(t) = 1$ and $x_{j'}(t) = 0, \forall j' \neq j \in \mathcal{J}$. In other words, the boundary solution is achieved when *all* workers in the population select a certain edge server j whereas no worker from the population selects other edge servers. For the latter, we have $x_j(t) = (0, 1), \forall j \in \mathcal{J}$. However, the boundary solution is not an ESS since a small perturbation results in a deviation and instability of the system states [35], [42]. Accordingly, we establish that the interior evolutionary equilibrium is the evolutionary stable strategy (ESS) in the following theorem:

Theorem 2. *The interior solution to the replicator dynamics, i.e., the interior evolutionary equilibrium $\mathbf{x}^*$, is the ESS.*

Proof. At the interior evolutionary equilibrium, we have $x_j^* N$ workers choosing the edge server j . Accordingly, the workers adopting this strategy derives the utility of $\frac{\frac{\zeta_j R_j}{N} W \mathbb{E}_{h_j} (\log_2 (1 + \frac{P_{h_j}}{W N_0}))}{S}$ where $\tilde{N} = x_j^* N$. Similarly, the utility of workers that select edge server j' is $\frac{\frac{\zeta_{j'} R_{j'}}{N} W \mathbb{E}_{h_{j'}} (\log_2 (1 + \frac{P_{h_{j'}}}{W N_0}))}{S}$ where $\hat{N} = x_{j'}^* N$. At the evolutionary equilibrium, the utilities derived from both strategies must be equivalent according to (4). Assuming that a single worker that selects cluster j churns and chooses j' , the revised utilities of the workers choosing edge server j and j' are now

$\frac{\alpha \frac{\zeta_j R_j}{N-1} W \mathbb{E}_{h_j} (\log_2 (1 + \frac{P_{h_j}}{W N_0}))}{S}$ and $\frac{\alpha \frac{\zeta_{j'} R_{j'}}{N+1} W \mathbb{E}_{h_{j'}} (\log_2 (1 + \frac{P_{h_{j'}}}{W N_0}))}{S}$ respectively. However, since

$$\frac{\alpha \frac{\zeta_{j'} R_{j'}}{N+1} W \mathbb{E}_{h_{j'}} (\log_2 (1 + \frac{P_{h_{j'}}}{W N_0}))}{S} < \frac{\alpha \frac{\zeta_j R_j}{N} W \mathbb{E}_{h_j} (\log_2 (1 + \frac{P_{h_j}}{W N_0}))}{S},$$

the relation implies that the worker does not have the incentive to switch his strategy in the first place as $\frac{\alpha \frac{\zeta_{j'} R_{j'}}{N+1} W \mathbb{E}_{h_{j'}} (\log_2 (1 + \frac{P_{h_{j'}}}{W N_0}))}{S} < \frac{\alpha \frac{\zeta_j R_j}{N} W \mathbb{E}_{h_j} (\log_2 (1 + \frac{P_{h_j}}{W N_0}))}{S}$. As such, the interior evolutionary equilibrium is a strict Nash equilibrium, which corresponds to the ESS [48]. $\square$

Theorem 3. *From an initial interior population state $\mathbf{x}(0)$ where $x_j(0) = (0, 1), \forall j \in \mathcal{J}$, the replicator dynamics is globally asymptotically stable and converges to the ESS.*

Proof. From (4), we rewrite the replicator dynamics in the vector form of $\dot{\mathbf{x}}_j = \mathbf{A} \mathbf{x} + \mathbf{b}$ where:

$$\mathbf{A} = -a \mathbf{I},$$

$$\text{where } a = \sum_{j=1}^J \frac{\frac{\zeta_j R_j}{N} \delta \alpha W \mathbb{E}_{h_j} \left(\log_2 \left(1 + \frac{P_{h_j}}{W N_0} \right) \right)}{S}, \text{ and}$$

$$\mathbf{b} = \left[\frac{\frac{\zeta_1 R_1}{N} \delta \alpha W \mathbb{E}_{h_1} \left(\log_2 \left(1 + \frac{P_{h_1}}{W N_0} \right) \right)}{S}, \dots, \frac{\frac{\zeta_j R_j}{N} \delta \alpha W \mathbb{E}_{h_j} \left(\log_2 \left(1 + \frac{P_{h_j}}{W N_0} \right) \right)}{S}, \dots, \frac{\frac{\zeta_J R_J}{N} \delta \alpha W \mathbb{E}_{h_J} \left(\log_2 \left(1 + \frac{P_{h_J}}{W N_0} \right) \right)}{S} \right]^T.$$

Note that $\mathbf{I}$ is a $J \times J$ identity matrix, and $\mathbf{A}$ is a $J \times J$ matrix. The characteristic equation of the replicator dynamics is given by:

$$\det(\kappa \mathbf{I} - \mathbf{A}) = (\kappa + a)^J = 0,$$

where $a > 0, \forall \zeta_j \in \zeta$. As such, $\mathbf{A}$ has J repeated negative eigenvalues, thus ensuring the global asymptotic stability of the ESS [49] under any bandwidth allocation control strategy adopted by the edge servers. $\square$

V. UPPER-LEVEL STACKELBERG DIFFERENTIAL GAME

In this section, we adopt the Stackelberg differential game to model the bandwidth allocation strategies of the edge servers, as well as the reward allocation scheme of the model owner. In Section V-A, we provide the game formulation and introduce the profit functions of the edge servers and model owner. In Section V-B, we derive the open-loop Stackelberg equilibrium as a solution to the game.

A. Problem Formulation

The two types of players and their respective strategies in the Stackelberg differential game are as follows:

- *Model owner*: As the leader of the game, the model owner devises the optimal reward to be accorded to the edge servers for facilitating the HFL process. The strategy of the model owner is the dynamic control of the reward $\beta(t)$, i.e., the payoff that the model owner accords to the edge server for *successfully* incorporating each worker in the HFL training.
- *Edge servers*: As the followers of the game, the edge servers devise an optimal bandwidth allocation control strategy in response to the reward scheme that the model owner sets. The strategy of the edge server is the dynamic control of $\zeta_j(t)$ where $\zeta_j(t) \in [0, 1]$.

Following [44], the model parameters are transmitted as packets in the uplink from the workers to the edge server. To detect packet transmission errors, the edge server conducts procedures such as the parity check and cyclic redundancy check (CRC) [44], [50]. The packet error rate for each uplink transmission [51] is given as follows:

$$q_j(\mathbf{x}(t), \zeta_j(t)) = r_j(\mathbf{x}(t), \zeta_j(t)) \mathbb{E}_{h_j} \left(1 - \exp \left(-\frac{v(WN_0)}{Ph_j} \right) \right), \quad (8)$$

where v is a waterfall threshold [51]. Accordingly, the probability of successful transmission is $(1 - q_j(\mathbf{x}(t), \zeta_j(t))) \in [0, 1]$. If a packet error is detected, the model parameters from the particular worker are discarded. For ease of notation, we express $q_j(\mathbf{x}(t), \zeta_j(t))$ and $q_j(t)$ interchangeably in this section.

The edge servers dynamically adapt their bandwidth allocation strategies in response to the dynamic population states $\mathbf{x}(t)$ and model owner reward $\beta(t)$. The objective of the edge server is to maximize its discounted cumulative payoff. The edge server optimal control problem is therefore formulated as:

$$\begin{aligned} \max_{\zeta_j(t)} \pi_j(\zeta_j(t), \mathbf{x}(t), \beta(t)) = & \int_0^\infty e^{-\rho t} \beta(t) (1 - q_j(t)) x_j(t) N - \theta_j \mathcal{C}(\zeta_j(t)) dt, \\ \text{s.t. } \dot{x}_j(t) = & \delta x_j(t) (u_j(B_j(t)) - \bar{u}(B_j(t))), \forall j \in \mathcal{J}, \\ & \mathbf{x}(0) = \bar{\mathbf{x}}, \end{aligned} \quad (9)$$

where ρ is the discount rate to translate future profits to current profits, e.g., $\rho = 1$ implies that the edge server values future and current profits equally whereas a lower ρ implies that the edge server values future profits less. Note that from (8), the edge server j will need to omit $q_j(t)x_j(t)N$ workers from the aggregation phase due to the erroneous transmission of model parameters. θ_j is the opportunity cost of bandwidth allocation, e.g., the bandwidth allocated by the edge server for the HFL process can be used to generate profits from the provision of other services, e.g., for provision of bandwidth for the closed subscriber group of small cell service providers [35]. Following [36], $\mathcal{C}(\cdot)$ is convex, i.e., $\mathcal{C}(\zeta_j(t)) = \zeta_j(t)^2$.

On the other hand, the model owner optimal control problem is formulated as:

$$\begin{aligned} \max_{\beta(t)} \pi(\beta(t), \zeta(t), \mathbf{x}(t)) = & \int_0^\infty e^{-\rho t} \sum_{j=1}^J \omega (\zeta_j(t) R_j) - \beta(t) (1 - q_j(t)) x_j(t) N \\ \text{s.t. } \dot{x}_j(t) = & \delta x_j(t) (u_j(B_j(t)) - \bar{u}(B_j(t))), \forall j \in \mathcal{J}, \\ & \mathbf{x}(0) = \bar{\mathbf{x}}, \end{aligned} \quad (10)$$

where $\omega > 0$ refers to the model owner preference for bandwidth contributed from the edge servers. Specifically, ω is larger if a model owner requires the learning task to be completed more urgently, e.g., when information freshness [52] and thus higher uplink rates are important for model training, or when more workers are required to train the model collaboratively, e.g., for location dependent tasks such as location-based recommender systems since each worker may only cover a small area of interest [53].

Clearly, the edge server and model owner optimal control problem are interdependent. As such, we subsequently derive the solutions to these problems by formulating the Stackelberg differential game.

B. Open-loop Stackelberg Equilibrium

In this section, we consider the open-loop Stackelberg equilibrium as a solution to the edge server and model owner optimal control problem [36]. As compared to the closed-loop equilibrium, the open-loop equilibrium can be derived without requiring any feedback information from the system. In other words, the outputs that result from the devised control strategies of the edge servers and model owner do not need to be observed when the optimal control strategies are derived. This removes the need for a central controller to govern the system, thus reducing the risk of a single point of failure and additional overheads. In addition, the self-organizing nature of the HFL network lowers the coordination and communication costs, both of which are bottlenecks of conventional FL system.

We first define the optimal control strategies as follows. For the edge servers, an optimal bandwidth allocation control path $\zeta_j^*(t)$ has to satisfy the following inequality condition:

$$\pi_j(\zeta_j^*(t), \mathbf{x}(t), \beta(t)) \geq \pi_j(\zeta_j(t), \mathbf{x}(t), \beta(t)). \quad (11)$$

For the model owner, an optimal reward allocation control strategy $\beta^*(t)$ has to satisfy the following inequality condition:

$$\pi(\beta^*(t), \zeta(t), \mathbf{x}(t)) \geq \pi(\beta(t), \zeta(t), \mathbf{x}(t)). \quad (12)$$

Accordingly, the open-loop Stackelberg equilibrium is characterized by the strategy profile $\{\beta^*(t), \zeta_j^*(t)\}$, in which the model owner reward allocation control strategy and edge server bandwidth allocation strategy maximizes their respective discounted profits, given each other's strategies. In order to obtain the open-loop equilibrium, we use the Pontryagin's maximum principle to derive the necessary conditions required to obtain the candidate optimal strategies [54]. In addition, the model owner is unable to observe the decisions or strategies

of the edge servers in an open-loop system. As such, the model owner devises an optimal reward scheme by taking into account the *expected best response* of the edge servers. Therefore, to analyze the Stackelberg differential equilibrium, we adopt the backward induction method.

We first define the Hamiltonian function of the edge server, i.e., the follower:

$$H_j(\zeta_j(t), \mathbf{x}(t), \beta(t), \Lambda_j(t), t) = \beta(1 - q_j(t))x_j(t)N - \theta_j\zeta_j(t)^2 + \Lambda_j(t)\delta x_j(t)(u_j(B_j(t)) - \bar{u}(B_j(t))), \quad (13)$$

where $\Lambda_j = [\lambda_{j,1}, \dots, \lambda_{j,J}]$ and $\lambda_{j,j}(t)$ refers to the co-state variable of edge server j associated with the population states in $\mathbf{x}(t)$. The maximized Hamiltonian function of the edge server is therefore defined by:

$$H_j^*(\mathbf{x}(t), \Lambda_j, t) = \max_{\zeta_j(t)} \{H_j(\zeta_j(t), \mathbf{x}(t), \beta(t), \Lambda_j(t), t) \mid \zeta_j(t) \in [0, 1]\}. \quad (14)$$

Given that the bandwidth control strategy of the edge server also has to maximize the Hamiltonian function, the candidate optimal strategies have to satisfy the following necessary conditions:

$$\frac{\partial H_j(\zeta_j(t), \mathbf{x}(t), \beta(t), \Lambda_j(t), t)}{\partial \zeta_j(t)} = 0, \quad (15)$$

$$\dot{\lambda}_{j,j}(t) = \rho\lambda_{j,j}(t) - \frac{\partial H_j^*(\mathbf{x}(t), \Lambda_j(t), t)}{\partial x_j(t)}. \quad (16)$$

Note that (16) refers to the adjoint equation that characterizes the dynamics of the co-state variable. The concavity of the Hamiltonian function with respect to $\zeta_j(t)$ ensures that a candidate optimal control path for ζ_j^* as a function of $\beta(t)$ can be uniquely determined as follows:

In simplifying (16), we have:

$$\dot{\lambda}_{j,j}(t) = \rho\lambda_{j,j}(t) - \beta(t)N + \frac{\lambda_{j,j}(t)\delta\alpha\zeta_j(t)R_j}{NS}W\mathbb{E}_{h_j}\left(\log_2\left(1 + \frac{Ph_j}{WN_0}\right)\right). \quad (18)$$

(17) and (18) characterize the best responses of the edge servers to the reward allocation control strategy of the model owner. To solve for the reward allocation control path, we consider the Hamiltonian function of the model owner, i.e., the leader in the Stackelberg differential game:

$$H(\beta(t), \mathbf{x}(t), \zeta(t), \sigma_j(t), \phi_j(t), \Lambda(t), t) = \sum_{j=1}^J \omega(\zeta_j(t)R_j) - \beta(t)(1 - q_j(t))x_j(t)N + \sum_{j=1}^J \sigma_j\delta x_j(t)(u_j(B_j(t)) - \bar{u}(B_j(t))) + \sum_{j=1}^J \phi_j(t)\dot{\Lambda}_j(t). \quad (19)$$

The model owner's profits are dependent on both the population states and bandwidth allocation control strategies of the edge servers. Consequentially, the Hamiltonian function of the

model owner include σ_j which refers to the co-state variable associated with each population state in $\mathbf{x}(t)$, and ϕ_j which refers to the co-state variable associated with each co-states of the edge servers in $\Lambda(t)$ and where $\phi_j = [\phi_1, \dots, \phi_J]$. The maximized Hamiltonian function of the model owner is therefore defined by:

$$H^*(\mathbf{x}(t), \sigma_j(t), \phi_j(t), t) = \max_{\beta(t)} H(\beta(t), \mathbf{x}(t), \zeta(t), \sigma_j(t), \phi_j(t), \Lambda(t), t) \mid \beta(t) > 0\}. \quad (20)$$

To maximize the Hamiltonian function in (19), the necessary conditions to devise the reward allocation control strategy are as follows:

$$\frac{\partial H(\beta(t), \mathbf{x}(t), \zeta(t), \sigma_j(t), \phi_j(t), \Lambda(t), t)}{\partial \beta(t)} = 0, \quad (21)$$

$$\dot{\sigma}_j(t) = \rho\sigma_j(t) - \frac{\partial H^*(\mathbf{x}(t), \sigma_j(t), \phi_j(t), t)}{\partial x_j(t)}, \quad (22)$$

$$\dot{\phi}_j(t) = \rho\phi_j(t) - \frac{\partial H^*(\mathbf{x}(t), \sigma_j(t), \phi_j(t), t)}{\partial \lambda_j(t)}. \quad (23)$$

Similarly, the concavity of the model owner's Hamiltonian function in (19) with respect to $\beta(t)$ ensures that a unique candidate optimal strategy $\beta^*(t)$ can be derived [35], [55]. By backward induction, i.e., substituting $\zeta_j^*(t)$ from (17) into $\beta_j^*(t)$, we can derive a function that does not explicitly show any of the control variable. As such, this reinforces the notion that the candidate Stackelberg differential equilibrium is open-loop in nature. Moreover, given the Pontryagin's maximum principle, the candidate strategy derived satisfies the necessary conditions for optimality. Next, since we are able to derive a unique solution to the replicator dynamics in (4), as well as that the Hamiltonian functions in (13) and (19) are concave and continuously differentiable with respect to $\mathbf{x}(t)$, the sufficient conditions for optimality of the Stackelberg equilibrium are also satisfied [35]. As such, the candidate strategy profile $\{\beta^*(t), \zeta_j^*(t)\}$ is the open-loop Stackelberg equilibrium.

The hierarchical game framework can therefore be implemented as follows in Algorithm 1. Following the procedure in [35], by substituting $\zeta_j^*(t)$ and $\beta_j^*(t)$ into (18), (22), and (23), we can obtain the optimal control path of the co-state $\dot{\lambda}_j(t)$ and $\dot{\sigma}_j(t)$, as well as the co-state of the co-states $\dot{\phi}_j(t)$. Subsequently, by solving the system of coupled differential equations, the Stackelberg differential equilibrium can be obtained, which gives us the optimal payment and bandwidth allocation control schemes. Then, the population state $\dot{\mathbf{x}}(t)$ can be derived from (4).

VI. PERFORMANCE EVALUATION

We present the performance evaluation of the evolutionary game based edge association and Stackelberg differential game based resource allocation in this section. Unless otherwise stated, the simulation parameters that we use are as follows in Table I, with reference to the studies in [?, [56], [57]. In Section VI-A, we first validate the existence, uniqueness, and stability of the evolutionary dynamics using numerical

$$\zeta_j^*(\beta(t), t) = \frac{\left(\lambda_{j,j}(t) \delta \frac{\alpha R_j}{NS} W \mathbb{E}_{h_j} \left(\log_2 \left(1 + \frac{Ph_j}{WN_0} \right) \right) - \beta(t) R_j \mathbb{E}_{h_j} \left(1 - \exp \left(-\frac{v(WN_0)}{Ph_j} \right) \right) \right)}{2\theta_j}. \quad (17)$$

Algorithm 1 Implementation of Hierarchical Game Framework

Input: Model owner, edge servers, workers attributes and system parameters

Output: $\beta^*(t)$, $\zeta_j^*(t)$, $x_j(t)$, $\forall j \in \mathcal{J}$

- 1: Initialize $\beta(0)$, $\zeta(0)$, $\mathbf{x}(0)$, $\lambda_j(0)$, $\sigma_j(0)$, $\phi_j(0)$, $\forall j \in \mathcal{J}$
 - 2: **for** $t \in \mathcal{T}$ **do**
 - 3: **Upper-Level Stackelberg Differential Game:**
 - 4: Compute the dynamics of the co-state variables $\dot{\lambda}_j(t)$, $\dot{\sigma}_j(t)$ and the co-state of the co-state $\dot{\phi}_j(t)$, $\forall j \in \mathcal{J}$.
 - 5: With $\lambda_j(t)$, $\sigma_j(t)$, and $\phi_j(t)$, derive $\beta^*(t)$ and $\zeta_j^*(t)$, $\forall j \in \mathcal{J}$.
 - 6: **Lower-Level Evolutionary Game:**
 - 7: Compute $u_j(B_j(t))$, $\forall j \in \mathcal{J}$ and $\bar{u}(B_j(t))$.
 - 8: **for** $j \in \mathcal{J}$ **do**
 - 9: Derive $\dot{x}_j(t) = \delta x_j(t) (u_j(B_j(t)) - \bar{u}(B_j(t)))$ and $x_j(t)$.
 - 10: **end for**
 - 11: **end for**
-

TABLE I: Simulation Parameters.

Parameters	Values
Total number of edge servers J	[2, 3]
Total number of worker populations M	[1, 3]
Number of workers in each population N	1000
Learning rate δ	[0.1, 1]
Noise power spectral density N_0	-174 dBm/Hz
Model parameter size S	10 MB
Bandwidth per resource block W	180kHz
Transmit power p_m	[0.5, 1] mW
Channel gain $h_{m,j}$	[2, 8] dBm
Resource blocks R_j	[100, 800]
Discount rate ρ	[0.5, 0.9]

experiments, in support of our discussion in Section IV. In consideration of the different sources of heterogeneity to be expected in practical HFL networks, we provide a sensitivity analysis in Section VI-B to study how the population states vary as a result. Finally in Section VI-C, we study the dynamic resource allocation under varying population states.

A. Validating the existence, uniqueness, and stability of the evolutionary dynamics

As an illustration to demonstrate the existence, uniqueness, and stability of the evolutionary equilibrium, we first consider the simplistic scenario of one population of workers and $J = 2$ edge servers. We initialize the population state to be $x_j(0) = 0.5$ for $j = \{1, 2\}$. In addition, we have $\zeta_1 R_1 = 500$ and $\zeta_2 R_2 = 900$, whereas $h_1 = h_2 = 8$ dBm. In other words, the populations are split evenly between the two edge servers at the beginning. In Fig. 2, we observe the evolutionary dynamics of the population shares x_1 and x_2 . We can observe that x_1 decreases steadily over time. The reason is that workers that have joined edge server 1 adapt their strategy to join

edge server 2 given that edge server 2 contributes more radio resource to facilitate the HFL process. Furthermore, we also observe the stability of the evolutionary equilibrium. After a certain number of iterations, the evolutionary equilibrium is reached and the workers from each population no longer have the incentive to deviate from their strategies.

In Fig. 3, we derive the phase plane of the replicator dynamics in (4). The parameters utilized are similar to that of our above discussion. We further consider varying initial population states represented by $\{x_1(0), x_2(0)\}$. For example, $\{x_1(0), x_2(0)\} = \{0.1, 0.9\}$ implies that 10% of the workers have chosen edge server 1, whereas the other 90% have chosen edge server 2. Then, we plot the trajectory of the evolutionary dynamics and present the evolution of the population shares $\{x_1(t), x_2(t)\}$. Clearly, despite beginning from varying initial conditions, the evolutionary equilibrium always converges to a unique interior solution, i.e., the ESS. In addition to the insights derived from Fig. 2, we are able to numerically validate the existence, uniqueness, and stability of the replicator dynamics.

B. Sensitivity Analysis

In consideration of the different sources of heterogeneity to be expected in practical HFL networks, we provide a sensitivity analysis in this section to consider different scenarios.

1) *Utility evolution involving multiple edge servers:* In Fig. 4, we consider the case of one population of workers and $J = 3$ edge servers. We initialize the population states such that the population of workers are split evenly among the three edge servers at the beginning. In addition, we have $\zeta_1 R_1 = 500$, $\zeta_2 R_2 = 900$, and $\zeta_3 R_3 = 600$. Similar to Fig. 2, we can observe that x_1 decreases steadily over time whereas x_2 increases due to the higher amount of radio resources that edge server $j = 2$ provides.

C. Dynamic Resource Allocation

To further understand the process behind the replicator dynamics, we consider the evolution of worker utilities during the strategy adaptation process. We plot the differences between the utilities derived from associating with each edge server and the average population utility in Fig. 5. At the beginning, we can observe that the utilities from associating with edge servers 1 and 3 are smaller than that of the average utility due to the larger amount of radio resource available for workers that associate with edge server 2. As such, following the replicator dynamics, we have more workers churning to associate with edge server 2. This causes less congestion in edge servers 1 and 3, and more congestion in edge server 2. As a result, the differences in utilities derived from associating with edge servers 1 and 3 becomes less negative, whereas the utility gains from associating with edge server 2 becomes less positive.

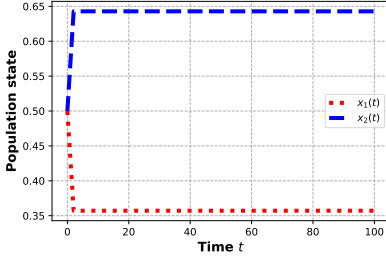


Fig. 2: Evolutionary dynamics of population state.

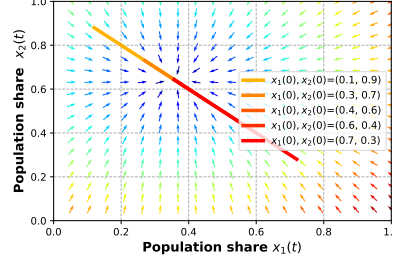


Fig. 3: Evolutionary dynamics under different initial population states.

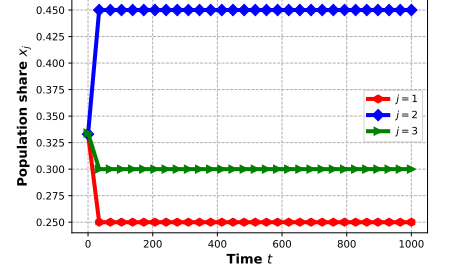


Fig. 4: Population state for population 1 with multiple populations.

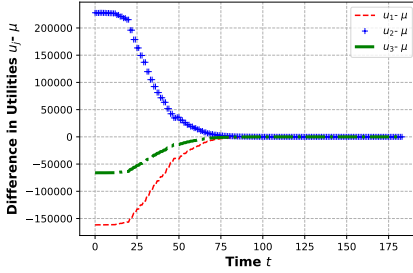


Fig. 5: Evolution of population utilities.

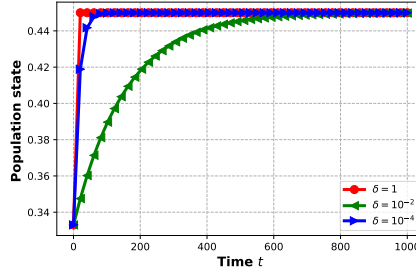


Fig. 6: Evolution of population state under different learning rates.

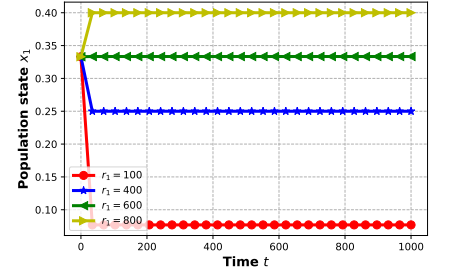


Fig. 7: Evolution of population state under heterogeneous resource constraints.

Eventually, at the evolutionary equilibrium, the utilities from associating with all three edge servers are equivalent and no workers have the incentive to change their strategies.

1) *Varying learning rates:* In Fig. 6, we vary the learning rate δ . Clearly, we can observe that the higher learning rates allow the evolutionary equilibrium to be reached faster and vice versa. A lower learning rate, however, does not affect the convergence of the equilibrium too adversely.

2) *Radio resource heterogeneity:* In Fig. 7, we vary the resource blocks that edge server 1 has, while fixing $R_j = 600$ for $j \in \{2, 3\}$. Then, we study how the variation in resource blocks for edge server 1 affects the population state x_1 , i.e., how the proportion of workers associating with edge server 1 changes with the respective variations in resource levels of cluster 1. As observed in Fig. 7, when the resource level of edge server 1 is relatively low, e.g., $R_1 = 100$, x_1 decreases sharply. The reason is that the edge server 1 has insufficient resources to support a large number of workers. The congestion leads to more workers choosing the other edge servers as a result. In contrast, when the resource levels of edge server 1 is increased, we observe that more workers eventually churn to edge server 1.

3) *Multiple heterogeneous populations:* Next, we consider the case of $M = 2$ populations of workers and $J = 3$ edge servers. In practice, to derive multiple populations, we may utilize clustering algorithms such as the k -means to segment the workers into heterogeneous populations based on their characteristics. In our case, the workers of population m has a channel gain of $h_{m,j}$ with edge server j . We denote $x_{m,j}(t)$ as the population share, i.e., proportion of population

m workers that associates with edge server j at time t , where $\sum_{j=1}^J x_{m,j}(t) = 1$ and $\sum_{m=1}^M \sum_{j=1}^J x_{m,j}(t) = M$.

We initialize the edge servers to have the same level of radio resource, i.e., $R_j = 400$ for $j \in \{1, 2, 3\}$. In addition, we have $h_{1,1} = h_{2,3} = 2\text{dBm}$, $h_{1,2} = h_{2,2} = 5\text{dBm}$, and $h_{1,3} = h_{2,1} = 8\text{dBm}$. In other words, $h_{1,1} < h_{1,2} < h_{1,3}$ whereas $h_{2,1} > h_{2,2} > h_{2,3}$. In Fig. 8, we chart the evolution of the population state for population 1, i.e., how $x_{1,j}$ changes with time where $j = \{1, 2, 3\}$. In Fig. 9, we chart the evolution of $x_{2,j}$. We can observe that $x_{1,1}$ falls with time for two reasons. Firstly, the cluster 1 has the least desirable channel gain for workers of population 1. As such, the workers switch their strategies to other clusters quickly. Secondly, the cluster 1 has the most desirable channel gain for workers of population 2. As such, the influx of workers from population 2 leads to competition of resources, which causes workers of population 1 to switch their strategies. The same argument thus applies for $x_{2,3}$. In contrast, the converse applies for $x_{1,3}$ and $x_{2,1}$. Next, we observe that the eventual population states at the equilibrium $x_{1,2}(T)$ and $x_{2,2}(T)$ are equivalent. The reason is that $h_{1,2} = h_{2,2}$, i.e., the workers of both populations rank the cluster similarly.

Finally, we observe how the evolutionary equilibrium is affected when we vary the channel conditions. We fix $h_{2,1} = 2\text{dBm}$ and $h_{2,j} = 8\text{dBm}$ for $j \in \{2, 3\}$ and vary $h_{1,j}$ between $[2\text{dBm}, 8\text{dBm}]$ for $\forall j \in \{1, 2, 3\}$. Then, we plot the variation of $x_{1,1}$ under the conditions of different $h_{1,j}$. As observed in Fig. 10, when $h_{1,j} = 2\text{dBm}$, we have most of the workers of population 1 churning to edge server 1. The reason is that

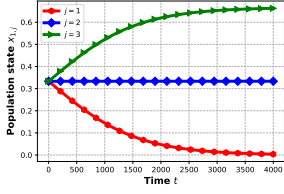


Fig. 8: Changes in evolutionary dynamics of $x_{1,j}$ under different channel conditions for population 1.

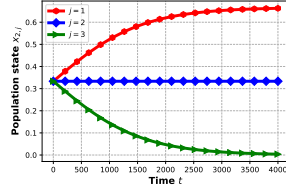


Fig. 9: Changes in evolutionary dynamics of $x_{2,j}$ under different channel conditions for population 2.

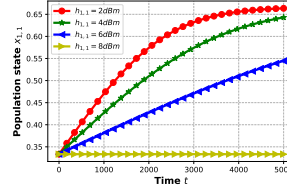


Fig. 10: Changes in evolutionary dynamics of $x_{1,1}$ under different channel conditions for population 1.

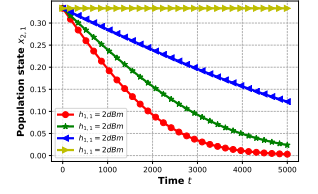


Fig. 11: Changes in evolutionary dynamics of $x_{2,1}$ under different channel conditions for population 1.

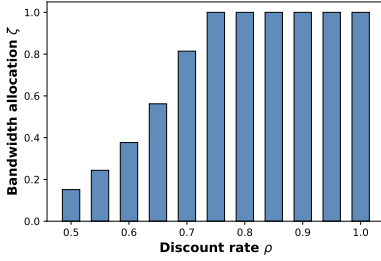


Fig. 12: Dynamic resource allocation under varying discount rates.

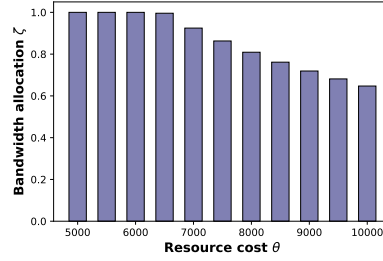


Fig. 13: Dynamic resource allocation under varying resource opportunity costs.

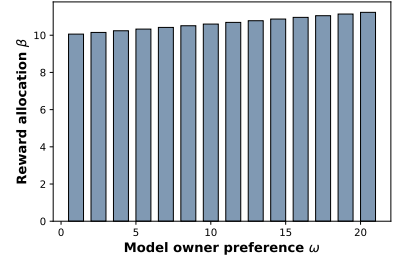


Fig. 14: Dynamic reward allocation under varying ω .

the workers of population 1 are indifferent between the three edge servers, whereas workers of population 2 prefer edge servers 2 and 3. As such, workers of population 1 will churn to edge server 1 which eventually has less competition for resources. In contrast, as we increase $h_{1,j}$, we observe that the population state of $x_{1,1}$ eventually decreases. The similar argument applies for the variations of population state $x_{2,1}$ in Fig. 11. In addition, for the case of multiple heterogeneous populations, we also observe that the convergence of the solution is achieved after many more iterations as compared to when a single population is involved, highlighting the additional time required for when this complex case is considered.

In this subsection, we study the Stackelberg differential game based dynamic resource allocation. With the parameters $R_1 = R_2 = R_3 = 600$ and the assumption of a fixed $\beta = 10$ for simplicity, we first vary the discount rate ρ to observe how the Stackelberg equilibrium for edge server resource allocation changes. In Fig. 12, we observe that when the discount rate is high, the bandwidth allocation $\zeta(t)$ remains at the equilibrium. In contrast, when discount rate is lower, the bandwidth allocation is lower at the equilibrium. The reason is that edge servers with a higher discount rate values the future stream of profits more than edge servers with lower discount rates. As a result, the edge servers with a higher discount rate tend to have sustainable contribution to the HFL training process in exchange for higher profits in the future time periods.

Apart from the discount rates, the resource cost also affects the bandwidth allocation strategy of the edge servers. In Fig. 13, we vary the opportunity cost of bandwidth allocation.

Besides, we keep $\rho = 0.8$ as a constant here. Evidently, when an edge server has lower opportunity cost of bandwidth allocation, the bandwidth allocation strategy of the edge server is higher. The reason is that edge servers with higher opportunity costs can channel their resources towards other profitable uses, thus reducing their contribution to the HFL process.

To study the reward allocation strategy of the model owner, we vary ω , which is the model owner preference for bandwidth contribution. Specifically, a model owner with higher ω may require more bandwidth, e.g., due to the fact that the HFL training process requires more workers to be involved, or because the model size is larger. As ω increases, we observe that the reward allocation strategy of the model owner also increases (Fig. 14). The reason is that the model owner is willing to pay more for the limited resources of the edge servers.

VII. CONCLUSION

In this paper, we first highlight the advantages and unique aspects of the HFL framework. Then, we propose a hierarchical game framework that utilizes an evolutionary game to model the dynamics of edge server association by the FL workers. Thereafter, a Stackelberg differential game is used to model the optimal bandwidth allocation and reward allocation strategies of the edge servers and model owner. Finally, through numerical experiments, we show that our proposed evolutionary and differential game based framework is able to model the HFL system dynamics under varying sources of heterogeneity to be expected in practical HFL networks. For future works, we may consider a case of multiple model

owners competing for the resources of edge servers. This may entail the use of self-revealing mechanisms such as auctions to appropriately price and allocate the resources in a HFL network.

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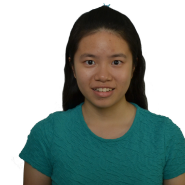
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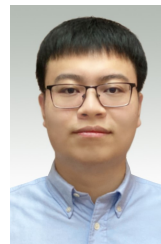
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